

PCI AI • FORMULA SHEET

The Project Controls Formula Sheet

Every formula the discipline runs on — and the mistake each one attracts

CPI • SPI • EAC • TCPI • ES • TF • EMV • CCC

PROJECT CONTROLS INSTITUTE GLOBAL

First edition • 2026 • AI proposes. The professional disposes.

What this is

Fifty-eight formulas across the whole of project controls: accounting, revenue recognition, budgeting, cost, earned value, forecasting, scheduling, risk, commercial and working capital.

Every one is drawn from the PCL-AI Body of Knowledge and carries the Knowledge Area where it is worked in full.

Against each, the error it attracts in practice — because a correct formula on a compromised input produces a confident, well-formatted, entirely wrong answer.

Save this one. It is the reference, not a highlight reel.

NOTATION · THE CLASH RULE

Two symbols. One page. Constant collision.

PV is **Planned Value** in earned value.

PV(x) is **present value** in discounting.

Both appear in the same monthly cost report.

Write "present value" in words, or **PV(x), whenever you discount.** This pair collides more often than any other in the discipline, and the Style Spine fixes it as a rule rather than a preference.

01

Accounting and reporting

Domains 1 and 2. Where project reality becomes a number the accounts will recognise.

The accounting core

FORMULA	MEANING	KA
$A = L + E$	The accounting equation	1.1.1
$\Sigma \text{ Debits} = \Sigma \text{ Credits}$	Double-entry invariant	1.1.3
$\text{Retained earnings} = \text{Opening} + \text{Income} - \text{Expenses} - \text{Distributions}$	Equity movement	1.1.1
$\text{Depreciation} = (\text{Cost} - \text{Residual}) \div \text{Useful life}$	Straight line	1.3.4
$\text{Carrying amount} = \text{Cost} - \text{Accumulated depreciation} - \text{Impairment}$	Net book value	1.3.4

A trial balance that balances proves arithmetic, not correctness. Both sides can be equally and identically wrong.

Provisions and the time value of an obligation

FORMULA	MEANING	KA
Expected value = Σ (probability_i $\times$ outcome_i)	Provision over a large population	1.4.3
Present value = Future amount $\div (1 + r)^n$	Discounting a provision	1.4.3

Expected value applies to a population. For a single obligation with one outcome, the most likely amount is the measure — using expected value there produces a number that will never be paid.

Revenue on a contract

FORMULA	MEANING	KA
PoC = Costs incurred ÷ Total estimated costs	Cost-to-cost percentage of completion	2.2.6
Cumulative revenue = PoC × Transaction price	Over-time revenue	2.2.6
Allocated price_i = Transaction price × (SSP_i ÷ Σ SSP)	Allocation to obligations	2.2.5
Capitalised borrowing cost = Weighted-avg qualifying expenditure × rate	IAS 23	2.4.4
Contract asset / (liability) = Revenue recognised – Amounts billed	Over- and under-billing	7.5

PoC is cumulative, and so is the revenue it produces.

Deduct what you have already recognised before booking the period. And uninstalled materials sitting in the numerator inflate progress that has not happened.

02

Budget, cost and variance

Domains 3 to 5. The baseline, the cost states, and the bridge from plan to actual.

Building the baseline

FORMULA	MEANING	KA
BAC = Σ control-account budgets + contingency	Cost baseline	3.1.4
Total authorised budget = BAC + Management reserve	Full authorisation	3.1.4
Analogous estimate = Past cost $\times$ (this driver $\div$ past driver)	Analogous estimating	3.2.2
Parametric estimate = Parameter $\times$ Rate	Parametric estimating	3.2.2

Management reserve sits outside BAC. Fold it in and you have quietly given yourself a contingency nobody approved and the baseline no longer measures anything.

DOMAIN 5 • COST

Cost behaviour and absorption

FORMULA	MEANING	KA
Total cost = Fixed + (Variable per unit × Volume)	Cost behaviour	5.1.1
OAR = Budgeted overhead ÷ Budgeted activity base	Overhead absorption rate	5.1.3
Over/(under) absorption = Absorbed – Incurred	Absorption variance	5.1.3

DOMAIN 5 • THE COST STATES

The one line that governs every index

$$\text{Cost to date} = \text{Actuals} + \text{Accruals}$$

Cost moves through three states — **commitment**, then **accrual**, then **actual**. Only the third has an invoice.

If **AC** carries invoiced cost alone, work has been performed that nobody has billed you for.

CPI overstates performance and EAC understates cost, and both errors are invisible in the report.

Reconcile goods-received-not-invoiced against the accrual booked before you compute anything in the next section.

DOMAIN 4 · VARIANCE

The variance bridge

FORMULA	MEANING	KA
Price variance = (Actual price – Standard price) × Actual quantity	Price effect	4.2.3
Quantity variance = (Actual qty – Standard qty) × Standard price	Quantity effect	4.2.3
Total variance = Price variance + Quantity variance	Reconciliation check	4.2.3

Note which multiplier belongs to which. Price variance takes **actual** quantity; quantity variance takes **standard** price. Swap them and the two will not sum to the total — which is exactly why the third line exists.

03

Earned value and forecasting

Domain 6, with Domain 9 for adaptive delivery. Where you are, and where you will end.

DOMAIN 6 · THE INDICES

Position

FORMULA	MEANING	KA
CV = EV – AC	Cost variance	6.2.1
SV = EV – PV	Schedule variance, in currency	6.2.1
CPI = EV ÷ AC	Cost efficiency to date	6.2.2
SPI = EV ÷ PV	Schedule efficiency in cost terms	6.2.2
% complete = EV ÷ BAC	Progress by value	6.1

SV and SPI are denominated in money, not time. At completion **EV = PV**, so both return to zero and 1.00 however late you finished. They are honest early and misleading late.

The EAC family

FORMULA	THE ASSUMPTION IT ENCODES	KA
$EAC = AC + ETC$	Identity — always true	6.3.1
$EAC = AC + (BAC - EV)$	The variance was a one-off	6.3.2
$EAC = BAC \div CPI$	Performance to date persists	6.3.2
$EAC = AC + (BAC - EV) \div CPI$	Same assumption, remaining-work form	6.3.2
$EAC = AC + (BAC - EV) \div (CPI \times SPI)$	Cost and schedule both persist	6.3.2
$VAC = BAC - EAC$	Variance at completion	6.3.4

The arithmetic is trivial. The assumption is the professional judgement — and it is the assumption a board will ask you to defend, not the division.

DECISION AID

Which EAC, and why

IF	THEN
The variance was a closed, one-off event	$AC + (BAC - EV)$
Performance to date will persist	$BAC \div CPI$
Cost and schedule pressure both persist	$AC + (BAC - EV) \div (CPI \times SPI)$
Remaining scope differs materially	Bottom-up ETC — indices do not apply

Publish the range and name the assumption against each. A single EAC quoted to the nearest thousand claims a precision the method cannot support.

DOMAIN 6 • TCPI

The check that exposes a wishful budget

FORMULA	MEANING	KA
$\text{TCPI (to BAC)} = \frac{\text{BAC} - \text{EV}}{\text{BAC} - \text{AC}}$	Efficiency needed to still hit budget	6.2.3
$\text{TCPI (to EAC)} = \frac{\text{BAC} - \text{EV}}{\text{EAC} - \text{AC}}$	Efficiency needed to hit a revised forecast	6.2.3

Set **TCPI** beside the **CPI** you have actually achieved. A **TCPI** far above it is asserting an efficiency gain nobody has demonstrated. And an identity worth knowing: when **EAC = BAC ÷ CPI**, **TCPI (to EAC)** equals **CPI** exactly.

DOMAIN 6 · EARNED SCHEDULE

Schedule performance, measured in time

FORMULA	MEANING	KA
$ES = M + \frac{(EV - PV_M)}{(PV_{M+1} - PV_M)}$	Interpolate between the cumulative-PV periods bracketing EV	6.4.3
$SV(t) = ES - AT$	Schedule variance in time	6.4.3
$SPI(t) = ES \div AT$	Time-based schedule efficiency	6.4.3

These do not converge at completion. Late in a project, earned schedule is the honest measure and the currency-based pair is not. Use cumulative planned value in the interpolation, never the period figure.

Earned value under adaptive delivery

FORMULA	MEANING	KA
% complete = Points completed ÷ Total planned points	AgileEVM progress	9.5.3
EV = % complete × BAC	Earned value from points	9.5.3
Velocity = Points completed ÷ Sprint	Delivery rate	9.3
Sprints remaining = Points remaining ÷ Velocity	Forecast to completion	9.3
Run rate = Team cost ÷ Sprint	Cost per sprint	9.5.2
Cycle time = WIP ÷ Throughput	Little's Law	9.4

Point inflation breaks the first two lines. If a story point in sprint 20 is smaller than in sprint 4, **% complete** rises without work being done. And use a rolling average velocity, never best-ever.

04

Schedule and risk

Domains 10 and 12. The network, its float, and the uncertainty priced into contingency.

Network and float

FORMULA	MEANING	KA
EF = ES + Duration	Forward pass	10.2
LS = LF – Duration	Backward pass	10.2
Total float = LS – ES = LF – EF	Slack without delaying the project	10.2.4
Free float = min(successor ES) – EF	Slack without delaying a successor	10.2.4
Critical path: TF = 0	The longest path	10.2.3

A free integrity check. Total float from the starts must equal total float from the finishes. If the two disagree, one of your passes is wrong — and constraints can manufacture a zero float that is not a critical path at all.

Duration under uncertainty

FORMULA	MEANING	KA
$tE = (O + 4M + P) \div 6$	PERT expected duration	10.1.4
$\sigma = (P - O) \div 6$	PERT standard deviation	10.1.4
$\sigma^2 \text{ path} = \sum \sigma^2 \text{ along the path}$	Path variance	10.3.4
$\text{Crash cost slope} = (\text{Crash} - \text{Normal cost}) \div (\text{Normal} - \text{Crash duration})$	Cost per period saved	10.3.1

Variances add. Standard deviations do not. Sum the σ^2 along the path and take the square root once, at the end. And crashing an activity that is not on the critical path spends money to save nothing.

Risk, and where contingency comes from

FORMULA	MEANING	KA
EMV = Probability × Impact	Expected monetary value	12.2.3
Total EMV = Σ (P_i × I_i)	Portfolio exposure	12.2.3
Contingency ≈ Σ EMV, or P80 – P50	Risk-based contingency	12.3.1

A percentage of budget is a habit, not an analysis. If contingency does not reconcile to quantified exposure, nobody can say what it is for or when it may be released.

05

Commercial and cash

Domains 7 and 11. The contract, the valuation and the working capital underneath.

Contract mechanics

FORMULA	MEANING	KA
Fee = Target fee + Contractor share × (Target cost – Actual cost)	CPIF incentive fee	7.1.3
Pain/gain = Share ratio × (Actual – Target)	Target-cost mechanism	7.1.4
LD exposure = LD rate × Days late	Liquidated damages	7.2.3
Remeasured value = Actual quantity × BoQ rate	Remeasurement	7.3.4

DOMAIN 7 · VALUATION

Getting paid

Amount due = $\Sigma(\% \text{ complete} \times \text{item amount}) - \text{Retention} - \text{Previous payments}$

Every interim application is this line. The failures are always in the last two terms.

Retention release is the one most often missed at the milestone that triggers it — and previous payments must be cumulative, or you will bill the same work twice and find out at final account.

Working capital

FORMULA	MEANING	KA
DSO = Receivables ÷ Daily revenue	Days sales outstanding	11.1.3
DIO = Inventory ÷ Daily COGS	Days inventory outstanding	11.A.1
DPO = Payables ÷ Daily COGS	Days payables outstanding	11.A.1
CCC = DSO + DIO – DPO	Cash conversion cycle	11.A.1
Cash freed ≈ Δ DSO × Daily revenue	Working capital released	11.A.1

DPO carries a minus sign. Paying suppliers later shortens your cash cycle and lengthens theirs — which is a commercial decision about the supply chain, not a treasury optimisation.

PROGRESS MEASUREMENT

Never accept this number

WORK PACKAGE CHARACTER	METHOD
Spans a single period	0/100
Two periods, discrete ends	50/50
Countable homogeneous output	Units complete
Long, with verifiable stages	Milestone weighting
No discrete output	Level of effort — never the default

Fix the method per work package before work starts.
And never accept a percentage complete supplied by the party being measured. That figure is a commercial position, not a measurement.

MOST MISAPPLIED · 1-5

The ten that cost the most

1. **CPI** computed on invoiced cost, with no accrual
2. **SV** read as a schedule measure late in a project
3. A single **EAC** presented as the forecast
4. **TCPI** never set beside the **CPI** achieved
5. Percentage complete supplied by the measured party

MOST MISAPPLIED · 6-10

And the other five

1. Standard deviations summed along a schedule path
2. **EMV** applied to a single binary event
3. Contingency set as a percentage of budget
4. Crashing an activity that is not on the critical path
5. Management reserve folded inside **BAC**

Every one of these produces a report that passes review. That is precisely why they are the expensive ones.

The Body of Knowledge is open

Every formula here is developed to full handbook depth in the PCL-AI Body of Knowledge — thirteen domains, sixty-one Knowledge Areas, with worked examples, figures and self-check exercises at the KA cited on each slide.

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